

HAMEENLINNA LAMMI PAPPILA, Finland

WMO: 027670

Lat: 61.0541N

Lon: 25.0385E

Elev: 125

StdP: 99.83

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -24.2	(c) -20.6	(d) -26.5	(e) 0.3	(f) -24.1	(g) -22.6	(h) 0.5	(i) -20.4	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.8	(c) 27.0	(d) 18.7	(e) 25.1	(f) 17.4	(g) 23.3	(h) 16.8	(i) 20.2	(j) 24.8	(k) 19.1	(l) 23.6	(m) 17.8	(n) 21.8	(o) N/A	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.6	13.6	22.2	17.3	12.5	21.1	16.1	11.6	19.5	58.6	24.9	54.7	23.6	50.7	21.9	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) N/A	(b) N/A	(c) N/A	(e) -26.2	(f) 29.0	(g) 4.4	(h) 2.5	(i) -29.3	(j) 30.7	(k) -31.9	(l) 32.2	(m) -34.4	(n) 33.6	(o) -37.6	(p) 35.3		
(4) N/A	(5) N/A	(6) N/A	(e) -26.1	(f) 21.2	(g) 4.3	(h) 1.9	(i) -29.1	(j) 22.6	(k) -31.6	(l) 23.7	(m) -34.0	(n) 24.8	(o) -37.2	(p) 26.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.8	-7.4	-5.5	-2.3	3.7	10.2	13.8	16.8	15.2	10.6	4.7	0.8	-3.2		
	DBStd	9.22	6.63	6.41	4.65	3.58	4.18	3.47	3.31	2.99	3.30	3.59	4.17	5.61		
	HDD10.0	2487	539	434	381	191	49	7	0	2	33	167	275	409		
	HDD18.3	4962	797	667	639	439	253	142	72	106	231	423	525	668		
	CDD10.0	604	0	0	0	3	56	120	210	162	51	3	0	0		
	CDD18.3	37	0	0	0	0	1	6	23	7	0	0	0	0		
	CDH23.3	385	0	0	0	0	31	72	219	63	0	0	0	0		
	CDH26.7	63	0	0	0	0	2	8	42	12	0	0	0	0		
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(15) Precipitation	PrecAvg	616	47	35	32	29	40	68	80	70	52	62	55	46		
	PrecMax	799	86	79	58	58	79	112	204	136	131	161	114	91		
	PrecMin	519	9	2	6	5	7	26	5	14	17	17	3	11		
	PrecStd	70	18	19	14	15	20	25	42	34	29	33	25	19		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.4	5.2	9.9	19.7	26.3	27.9	29.7	28.1	21.0	13.9	9.7	7.9		
		MCWB	2.7	2.7	4.9	10.8	15.0	18.6	20.0	19.5	16.1	11.8	8.9	7.4		
	2%	DB	2.4	3.1	6.7	15.9	23.8	25.4	27.9	24.8	18.9	12.0	8.7	5.2		
		MCWB	2.1	2.2	3.3	8.4	14.6	16.7	19.5	17.9	15.1	10.3	8.1	4.5		
	5%	DB	1.4	2.0	4.7	13.2	21.3	23.4	26.0	22.9	17.2	11.0	7.1	3.7		
		MCWB	1.1	1.5	2.4	7.3	13.5	16.1	18.8	17.0	14.0	10.0	6.6	3.1		
	10%	DB	0.4	1.2	3.1	10.4	18.9	21.2	23.9	21.0	15.7	10.0	5.8	2.5		
		MCWB	0.2	0.8	1.4	5.7	12.2	15.0	18.1	16.2	13.5	9.0	5.4	2.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.1	3.2	5.6	11.4	17.2	20.3	21.6	21.2	16.9	12.6	9.2	7.5		
		MCDB	3.3	4.0	8.3	17.6	22.6	24.2	26.5	26.1	19.5	13.6	9.7	8.0		
	2%	WB	2.1	2.3	3.9	9.6	15.7	18.2	20.6	19.2	15.8	11.2	8.1	4.7		
		MCDB	2.3	3.0	6.3	15.1	20.9	23.1	25.6	22.7	18.0	11.7	8.7	5.1		
	5%	WB	1.2	1.6	2.8	7.7	14.5	16.9	19.8	17.9	14.8	10.3	6.6	3.2		
		MCDB	1.4	2.1	4.2	11.8	19.8	22.1	24.5	21.7	16.6	10.9	7.0	3.6		
	10%	WB	0.3	0.8	1.8	6.4	13.0	15.7	18.7	16.9	13.8	9.1	5.6	2.2		
		MCDB	0.5	1.2	2.9	9.9	18.1	20.3	23.2	19.9	15.3	9.8	5.9	2.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.3	5.4	7.4	9.4	11.2	10.1	9.8	9.5	7.6	5.6	3.6	3.9		
		MCDBR	3.3	4.4	8.6	14.7	15.9	14.0	12.8	12.6	9.9	6.7	4.4	4.5		
		MCWBR	3.2	3.9	5.8	8.2	7.8	6.9	5.9	6.6	6.4	5.2	4.3	4.3		
	5% WB	MCDBR	3.4	4.1	7.2	12.6	13.3	11.9	11.6	10.7	8.8	5.9	4.3	4.3		
		MCWBR	3.3	3.8	5.3	7.6	7.1	6.3	5.7	6.0	6.0	5.2	4.2	4.2		
(40) Clear-Sky Solar Irradiance	taub		0.257	0.298	0.324	0.357	0.354	0.351	0.373	0.362	0.337	0.322	0.282	0.368		
	taud		2.095	2.166	2.241	2.365	2.430	2.462	2.415	2.442	2.507	2.407	2.191	2.617		
	Ebn at Noon		462	670	789	837	870	878	845	826	786	654	431	315		
	Edh at Noon		40	70	92	99	101	99	102	92	72	57	37	25		
(44) All-Sky Solar Radiation	RadAvg		0.23	0.80	2.28	3.75	5.06	5.29	5.12	3.96	2.35	1.01	0.28	0.13		
	RadStd		0.04	0.18	0.29	0.39	0.56	0.42	0.50	0.44	0.31	0.17	0.05	0.02		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (60 neighbors)		+0.48	N/A	N/A	N/A	N/A	N/A	-145	N/A	N/A	N/A

Nomenclature: See separate page